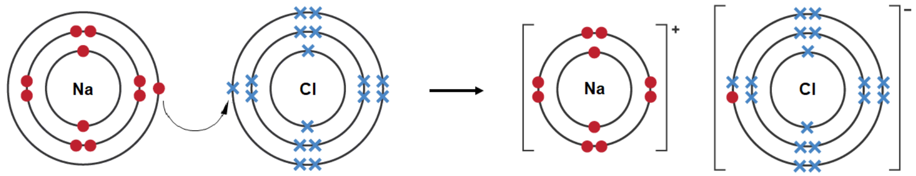
SECTION B

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
9	(a)	(i)	I	each carbon atom in diamond has four (strong) covalent bonds at tetrahedral angles / in tetrahedral structure (1) each carbon atom in graphite has three (strong) covalent bonds within the same plane / forming hexagons in planes (1) if no other credit award (1) if both statements partially correct	2			2		
			II	credit property common to both and corresponding explanation both have high melting points (1) due to giant molecular structure of strong covalent bonds / because a large amount of energy is needed to break strong covalent bonds (1) or both are insoluble in water (1) because they are non-polar (1) credit different property and corresponding explanation graphite is soft but diamond is hard (1) because graphite has weak forces between planes (1) or graphite conducts electricity but diamond does not (1) because graphite has delocalised electrons along the planes (1)	4			4		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(ii)	I	 electron transfer and correct electronic structure of ions (1) charge on both ions (1)		2		2		
			II	Na^+ **ion** smaller than Cs^+ **ion** (so fewer Cl^- ions can pack around it)	1			1		
	(b)	(i)		yellow (1) $Ag^+(aq) + I^-(aq) \rightarrow AgI(s)$ (1) ignore state symbols	1	1		2		1
		(ii)		iodine (causes the brown colouration) (1) (bromine is an) oxidising agent (1)	1	1		2		1
	(c)			add any suitable sulfate solution e.g. sodium sulfate (1) no change with Mg^{2+} and white precipitate with Ba^{2+} (1) or add sodium hydroxide solution (1) white precipitate with Mg^{2+} and no change with Ba^{2+} (1)	1		1	2		2
				Question 9 total	**10**	**4**	**1**	**15**	**0**	**4**